Parametric, Polar and Vector-Valued Calculus — Diagnosing Parameter, Polar-Bound and Speed-versus-Velocity Errors

AP Calculus BC

Every problem below contains work that is arithmetically clean and conceptually wrong. Three faults recur:

- **Parameter:** $\dfrac{dy}{dx} = \dfrac{dy/dt}{dx/dt}$, evaluated at the t that *reaches* the point — never at the point's own coordinate.
- **Polar bounds:** area is $\frac{1}{2}\displaystyle\int r^2\,d\theta$ taken over the θ-interval that traces the region *once*. Find where $r = 0$ before choosing limits.
- **Speed vs velocity:** velocity is the vector $\langle x'(t),\, y'(t)\rangle$; speed is its magnitude $\sqrt{x'(t)^2 + y'(t)^2}$, and a speed is never negative.
- For each problem: name the error in words, then show the corrected work.

1. Find and fix. A curve is given by $x(t) = t^2 + 2t$ and $y(t) = t^3 - 4t$. Ana wants $\dfrac{dy}{dx}$ at $t = 1$ and computes $\dfrac{dx/dt}{dy/dt}$, obtaining -4.

Name Ana's error in one sentence, then compute $\dfrac{dy}{dx}$ at $t = 1$.

Answer: ____________________

2. Find and fix. A particle moves along $x(t) = t^2 - 3t,\ \ y(t) = t^3$. Bo reports the speed at $t = 2$ as 13, having added $x'(2)$ and $y'(2)$.

Say what quantity Bo's number actually describes, then find the speed at $t = 2$, rounded to the nearest hundredth.

Answer: ________________

3. Find and fix. The rose $r = 4\cos(3\theta)$ has three petals. Cal wants the area of *one* petal, integrates $\frac{1}{2}r^2$ from 0 to 2π, and reports 25.13.

(a) Solve $r = 0$ to find the two consecutive θ-values that bound one petal.

(b) Using those limits, compute the exact area of one petal. Explain in one sentence what Cal's limits actually measured.

Answer: ________________

4. Find and fix. A particle's position is $x(t) = t^2,\ y(t) = \frac{2}{3}t^3$ for $0 \le t \le 1$. Dev finds the total distance travelled by integrating $x'(t)$ and $y'(t)$ separately and adding the two results, obtaining 1.667.

Write the correct integral for total distance, evaluate it exactly, and say in one sentence why Dev's method cannot give a distance.

Answer: ____________________

5. Find and fix. The curve $x(t) = t^2 + 1,\ y(t) = t^3 - 3t$ passes through the point $(2, -2)$. Elle finds the slope of the tangent there by substituting $t = 2$, "because $x = 2$", and reports 2.25.

(a) Find every t with $x(t) = 2$, and say which one reaches the point $(2, -2)$.

(b) Compute the correct slope $\dfrac{dy}{dx}$ at that point.

Answer: ____________________

6. Find and fix. A limaçon is given by $r = 1 + 2\cos\theta$. Ravi is asked for every θ in $[0, 2\pi)$ at which the curve passes through the pole. He writes $\cos\theta = \frac{1}{2}$ and answers $\theta = \frac{\pi}{3}$.
Ravi solved the wrong equation. Set up the correct one and give every solution in $[0, 2\pi)$, in exact form.

Answer: ____________________

7. Find and fix. A particle follows the vector-valued path $\mathbf{r}(t) = \langle \sin 2t,\ \cos t \rangle$. Nia computes the speed at $t = \frac{\pi}{3}$ by adding the two components of $\mathbf{r}'(t)$, and reports -1.87.
Give one reason Nia's answer must be wrong before doing any arithmetic. Then compute the speed at $t = \frac{\pi}{3}$, rounded to the nearest hundredth.

Answer: ____________________

8. Explain.

(a) Distinguish, in your own words, the velocity vector, the speed, and the slope $\dfrac{dy}{dx}$ for a parametric curve. Say which are vectors, which are scalars, and what each one tells you about the motion.
(b) Explain why a component of velocity may be negative but a speed never can.
(c) Explain why integrating $\frac{1}{2}r^2$ over a full turn overcounts the area of the rose in problem 3, and state a habit for choosing polar limits.